

Set up

In [1]: `!pip install prince`

Collecting prince

Downloading prince-0.16.0-py3-none-any.whl.metadata (641 bytes)

Requirement already satisfied: altair<6.0.0,>=5.0.0 in /opt/conda/lib/python3.11/site-packages (from prince) (5.4.1)

Requirement already satisfied: pandas<3.0.0,>=2.2.0 in /home/jovyan/.local/lib/python3.11/site-packages (from prince) (2.2.3)

Requirement already satisfied: scikit-learn<2.0.0,>=1.5.1 in /opt/conda/lib/python3.11/site-packages (from prince) (1.5.2)

Requirement already satisfied: jinja2 in /opt/conda/lib/python3.11/site-packages (from altair<6.0.0,>=5.0.0->prince) (3.1.4)

Requirement already satisfied: jsonschema>=3.0 in /opt/conda/lib/python3.11/site-packages (from altair<6.0.0,>=5.0.0->prince) (4.23.0)

Requirement already satisfied: narwhals>=1.5.2 in /opt/conda/lib/python3.11/site-packages (from altair<6.0.0,>=5.0.0->prince) (1.8.1)

Requirement already satisfied: packaging in /opt/conda/lib/python3.11/site-packages (from altair<6.0.0,>=5.0.0->prince) (24.1)

Requirement already satisfied: typing-extensions>=4.10.0 in /opt/conda/lib/python3.11/site-packages (from altair<6.0.0,>=5.0.0->prince) (4.12.2)

Requirement already satisfied: numpy>=1.23.2 in /home/jovyan/.local/lib/python3.11/site-packages (from pandas<3.0.0,>=2.2.0->prince) (2.0.2)

Requirement already satisfied: python-dateutil>=2.8.2 in /opt/conda/lib/python3.11/site-packages (from pandas<3.0.0,>=2.2.0->prince) (2.9.0)

Requirement already satisfied: pytz>=2020.1 in /opt/conda/lib/python3.11/site-packages (from pandas<3.0.0,>=2.2.0->prince) (2024.2)

Requirement already satisfied: tzdata>=2022.7 in /opt/conda/lib/python3.11/site-packages (from pandas<3.0.0,>=2.2.0->prince) (2024.1)

Requirement already satisfied: scipy>=1.6.0 in /home/jovyan/.local/lib/python3.11/site-packages (from scikit-learn<2.0.0,>=1.5.1->prince) (1.13.1)

Requirement already satisfied: joblib>=1.2.0 in /home/jovyan/.local/lib/python3.11/site-packages (from scikit-learn<2.0.0,>=1.5.1->prince) (1.4.2)

Requirement already satisfied: threadpoolctl>=3.1.0 in /opt/conda/lib/python3.11/site-packages (from scikit-learn<2.0.0,>=1.5.1->prince) (3.5.0)

Requirement already satisfied: attrs>=22.2.0 in /opt/conda/lib/python3.11/site-packages (from jsonschema>=3.0->altair<6.0.0,>=5.0.0->prince) (24.2.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /opt/conda/lib/python3.11/site-packages (from jsonschema>=3.0->altair<6.0.0,>=5.0.0->prince) (2023.12.1)

Requirement already satisfied: referencing>=0.28.4 in /opt/conda/lib/python3.11/site-packages (from jsonschema>=3.0->altair<6.0.0,>=5.0.0->prince) (0.35.1)

Requirement already satisfied: rpds-py>=0.7.1 in /opt/conda/lib/python3.11/site-packages (from jsonschema>=3.0->altair<6.0.0,>=5.0.0->prince) (0.20.0)

Requirement already satisfied: six>=1.5 in /opt/conda/lib/python3.11/site-packages (from python-dateutil>=2.8.2->pandas<3.0.0,>=2.2.0->prince) (1.16.0)

Requirement already satisfied: MarkupSafe>=2.0 in /opt/conda/lib/python3.11/site-packages (from jinja2->altair<6.0.0,>=5.0.0->prince) (2.1.5)

Downloading prince-0.16.0-py3-none-any.whl (417 kB)

Installing collected packages: prince

Successfully installed prince-0.16.0

In [1]: `import os
import re
import sys
import numpy as np`

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

import plotly.graph_objects as go
from mpl_toolkits.mplot3d import Axes3D

pd.set_option('display.max_columns', None)

import importlib
def reload_module(module):
    importlib.reload(module)

# Specify the directory containing the .py modules
module_dir = "modules"

# Add the directory to the Python path
sys.path.append(os.path.abspath(module_dir))
```

CRLA module

```
In [2]: # See crla.py for the raw code
from crla_v2 import CRLAv2 as CRLA
crla_analyzer = CRLA()
```

```
In [3]: help(crla_analyzer)
```

Help on CRLAV2 in module crla_v2 object:

```
class CRLAV2(builtins.object)
|   Classroom Reading Level Assessment (CRLA) data analysis class.
|
|   This class handles the processing, analysis, and visualization of
|   reading assessment data for Grades 1-3, comparing beginning of school year (BoSY)
|   and end of school year (EoSy) results to measure educational progress.
|
|   Methods defined here:
|
|   __init__(self)
|       Initialize the CRLA class.
|
|   analyze_reading_performance(self, processed_tables, combined_raw_data, combined_percentage_data, custom_weights=None)
|       Comprehensive analysis of reading performance with data validation.
|
|       Parameters
|       -----
|       processed_tables : dict
|           Dictionary of processed tables from calculate_raw_and_percentage_data
|       combined_raw_data : dict
|           Dictionary containing combined raw count data
|       combined_percentage_data : dict
|           Dictionary containing combined percentage data
|       custom_weights : dict, optional
|           Custom weights for each reading profile category
|
|       Returns
|       -----
|       dict
|           Dictionary containing analysis results
|
|   calculate_performance_score(self, school_data, validation=None, custom_weights=None)
|       Calculate a weighted performance score that works with incomplete data.
|
|       Parameters
|       -----
|       school_data : pandas.DataFrame
|           DataFrame containing percentages of students in each reading category
|       validation : pandas.DataFrame, optional
|           Validation data to identify problematic schools
|       custom_weights : dict, optional
|           Custom weights for each reading profile category
|
|       Returns
|       -----
|       pandas.Series
|           Series containing performance scores for each school
|
|   calculate_progress_score(self, bosy_data, eosy_data, validation, custom_weights=None)
|       Calculate progress score only for schools with valid data in both periods.
|
|       Parameters
|       -----
```

```

|     bosy_data : pandas.DataFrame
|         Beginning of School Year data
|     eosy_data : pandas.DataFrame
|         End of School Year data
|     validation : pandas.DataFrame
|         Validation data to identify which schools have valid data
|     custom_weights : dict, optional
|         Custom weights for each reading profile category
|
|     Returns
|     -----
|     pandas.Series
|         Series containing progress scores for valid schools
|
| calculate_raw_and_percentage_data(self, grade_tables, groupby_column='School
ID')
|     Calculate raw counts and percentages of students in each reading profile
category.
|
|     Parameters
|     -----
|     grade_tables : dict
|         Dictionary of grade-specific data tables from create_grade_data_table
s
|     groupby_column : str, optional
|         Column to use for grouping (e.g., "School ID", "Region")
|
|     Returns
|     -----
|     dict
|         Dictionary containing processed tables for each grade with raw counts
and percentages
|
| combine_grade_data(self, bosy_dataframes, eosy_dataframes, school_identifier
s)
|     Combine data from all grades into a single dataframe for each period.
|     Can be used for both raw counts or percentages.
|
|     Parameters
|     -----
|     bosy_dataframes : list
|         List of BoSY dataframes for G1, G2, G3
|     eosy_dataframes : list
|         List of EoSY dataframes for G1, G2, G3
|     school_identifiers : pandas.DataFrame
|         Table of school identifiers
|
|     Returns
|     -----
|     dict
|         Dictionary containing 'BoSY' and 'EoSY' dataframes with combined data
|
|     create_grade_data_tables(self, assessment_dataframes, school_identifiers, gra
de_columns_dict, regex_patterns=['(^G1) (\\b.*\\w$)', '(^G2) (MT|Fil) (\\b.*$)',
'(^G3) (MT|Fil|Eng) (\\b.*$)'])
|         Create grade-specific data tables from assessment data.
|
|     Parameters
|     -----
|     assessment_dataframes : list

```

```

|         List containing [bosy_dataframe, eosy_dataframe]
| school_identifiers : pandas.DataFrame
|         Table of school identifiers created by create_school_identifier_table
| grade_columns_dict : dict
|         Dictionary mapping grade levels to their data columns
| regex_patterns : list, optional
|         List of regex patterns to extract components from column headers for
each grade
|
| Returns
| -----
| dict
|         Dictionary mapping grade levels to their processed data tables
|
| create_school_identifier_table(self, identifier_columns, assessment_dataframe
s)
|     Create a unified table of school identifiers from both BoSY and EoSY dat
a.
|
| Parameters
| -----
| identifier_columns : list
|         List of column names that identify schools
| assessment_dataframes : list
|         List containing [bosy_dataframe, eosy_dataframe]
|
| Returns
| -----
| pandas.DataFrame
|         Table of school identifiers with School ID as index
|
| extract_column_structures(self, assessment_dataframe)
|     Extract identification columns and grade-specific data columns from asses
sment data.
|
| Parameters
| -----
| assessment_dataframe : pandas.DataFrame
|         Assessment data to extract columns from
|
| Returns
| -----
| tuple
|     (identifier_columns, grade_columns_dict) where:
|     - identifier_columns: list of column names that identify schools
|     - grade_columns_dict: dictionary mapping grade levels to their data c
olumns
|
| extract_data_by_period(self, processed_tables, data_type='percentages')
|     Extract data of specified type from processed tables for both periods.
|
| Parameters
| -----
| processed_tables : dict
|         Dictionary of processed tables from calculate_raw_and_percentage_data
| data_type : str, optional
|         Type of data to extract: 'raw' for raw counts, 'percentages' for perc
entages
|
| Returns

```

```

|         tuple
|         (bosy_dataframes, eosy_dataframes) containing lists of dataframes for
each grade
|
|     get_pca_components_info(self)
|         Create a DataFrame of PCA components with proper column names,
|         showing the loadings of each feature on the principal components.
|
|         Returns
|         -----
|         pandas.DataFrame
|             DataFrame containing PCA components with feature names as columns
|
|     load_assessment_data(self, directory_path=None, bosy_filename=None, eosy_file
name=None)
|         Load Beginning of School Year (BoSY) and End of School Year (EoS
Y) assess
ment data.
|
|         Parameters
|         -----
|         directory_path : str, optional
|             Path to the directory containing the data files
|         bosy_filename : str, optional
|             Filename for the Beginning of School Year data
|         eosy_filename : str, optional
|             Filename for the End of School Year data
|
|         Returns
|         -----
|         tuple
|             (bosy_dataframe, eosy_dataframe) containing the loaded assessment dat
a
|
|     optimize_weights_from_pca(self, combined_percentage_data=None, validation=Non
e, invert=False)
|         Derive optimal weights from PCA loadings on the first principal componen
t,
|         using only schools that are valid for progress analysis.
|
|         Parameters
|         -----
|         combined_percentage_data : dict, optional
|             Dictionary containing 'BoSY' and 'EoS'Y' dataframes with percentage da
ta.
|             If None, uses the stored self.combined_percentage_data.
|         validation : pandas.DataFrame, optional
|             Validation data containing the 'valid_for_progress' column.
|             If None, uses the stored self.validation_data.
|         invert : bool, optional
|             If True, invert the direction of weights for all categories except 'G
rade Level'
|             when 'Developing' has lower loading than 'Lower Emergent'.
|             If False, use the PCA loadings as they are with no inversion.
|
|         Returns
|         -----
|         dict
|             Dictionary of optimized weights for each reading profile category

```

```

| validate_assessment_data(self, bosy_data, eosy_data, raw_count_data=None, thr
eshold=0.25)
|     Validate school data to identify problematic cases for analysis.
|
|     Parameters
|     -----
|     bosy_data : pandas.DataFrame
|         Beginning of School Year percentage data
|     eosy_data : pandas.DataFrame
|         End of School Year percentage data
|     raw_count_data : dict, optional
|         Dictionary containing raw count data for both periods
|     threshold : float, optional
|         Maximum acceptable difference in student counts (as a proportion)
|
|     Returns
|     -----
|     pandas.DataFrame
|         DataFrame with validation flags for each school
|
|     -----
|     Data descriptors defined here:
|
|     __dict__
|         dictionary for instance variables (if defined)
|
|     __weakref__
|         list of weak references to the object (if defined)

```

Load and preprocessing

```

In [4]: %%time
# Load data from raw .CSV files of CRLA SY24-25 BoSY and EoSY
bosy_df, eosy_df = crla_analyzer.load_assessment_data(
    directory_path = 'data'
)

```

CPU times: user 464 ms, sys: 705 ms, total: 1.17 s
Wall time: 1.44 s

```

In [5]: %%time
# GENERAL NOTE: tables refer to stand-alone dataframes
# Load relevant variables for generating tables
identifier_columns, grade_columns_dict = crla_analyzer.extract_column_structures

```

CPU times: user 27.5 ms, sys: 314 µs, total: 27.8 ms
Wall time: 24.7 ms

```

In [6]: %%time
# Generate table of unique schools and their identifiers
school_identifiers = crla_analyzer.create_school_identifier_table(
    identifier_columns,
    assessment_dataframes=[bosy_df, eosy_df]
)

```

CPU times: user 40.2 ms, sys: 9.76 ms, total: 50 ms
Wall time: 39.9 ms

```
In [7]: school_identifiers.head()
```

```
Out[7]:
```

	Region	Division	District	School Name
School ID				
131240	BARMM	Cotabato City	Cotabato City District I	Cotabato City Central Pilot School
131244	BARMM	Cotabato City	Cotabato City District II	Sero Central School
131248	BARMM	Cotabato City	Cotabato City District III	L.R. Sebastian ES
131252	BARMM	Cotabato City	Cotabato City District III	Notre Dame Village Central ES
209504	BARMM	Cotabato City	Cotabato City District III	Mandanas Elementary School

```
In [8]: %%time
# Generate per grade level their corresponding concatenated BoSY & EoSY tables

grade_tables = crla_analyzer.create_grade_data_tables(
    [bosy_df, eosity_df],
    school_identifiers,
    grade_columns_dict
)
```

CPU times: user 12.1 s, sys: 1.81 s, total: 13.9 s
Wall time: 13.9 s

```
In [9]: %%time

# Transform BoSY and EoSY grade level tables to Long dataframes containing the c
# that fall under the domains/subdomains language and/or reading profiles. Coun
# count and percentage from total assessed in the domains/subdomains

processed_tables = crla_analyzer.calculate_raw_and_percentage_data(
    grade_tables,
    groupby_column = "School ID"
)
```

CPU times: user 7min 4s, sys: 161 ms, total: 7min 4s
Wall time: 7min 4s

In raw count

```
In [10]: %%time
bosy_raw_df, eosity_raw_df = crla_analyzer.extract_data_by_period(
    processed_tables,
    data_type='raw'
)
```

CPU times: user 8 µs, sys: 0 ns, total: 8 µs
Wall time: 10.7 µs

```
In [11]: %%time
combined_raw_df = crla_analyzer.combine_grade_data(
    bosal_raw_df,
    eosy_raw_df,
    school_identifiers
)
```

CPU times: user 39.9 ms, sys: 10 ms, total: 49.9 ms
Wall time: 48.9 ms

```
In [12]: combined_raw_df.keys()
```

```
Out[12]: dict_keys(['BoSY', 'EoSY'])
```

```
In [13]: combined_raw_df['BoSY'].head()
```

```
Out[13]:
```

	School Name	Region	Division	District	G1_Developing	G1_Grade Level	G1_Higher Emergent	G1_Emergent
School ID								
100000	Balayong Elementary School	Region I	San Carlos City	San Carlos City District I	9.0	4.0	7.0	
100001	Apaleng-Libtong ES	Region I	Ilocos Norte	Bacarra I	5.0	0.0	0.0	
100002	Bacarra CES	Region I	Ilocos Norte	Bacarra I	7.0	0.0	3.0	
100003	Buyon ES	Region I	Ilocos Norte	Bacarra I	0.0	3.0	0.0	
100004	Ganagan Elementary School	Region I	Ilocos Norte	Bacarra I	3.0	0.0	0.0	

```
In [14]: combined_raw_df['EoSY'].head()
```

Out[14]:

School ID	School Name	Region	Division	District	G1_Developing	G1_Grade Level	G1_Higher Emergent	G1_Emergent
100000	Balayong Elementary School	Region I	San Carlos City	San Carlos City District I	4.0	14.0	1.0	
100001	Apaleng-Libtong ES	Region I	Ilocos Norte	Bacarra I	1.0	6.0	0.0	
100002	Bacarra CES	Region I	Ilocos Norte	Bacarra I	8.0	23.0	2.0	
100003	Buyon ES	Region I	Ilocos Norte	Bacarra I	0.0	7.0	0.0	
100004	Ganagan Elementary School	Region I	Ilocos Norte	Bacarra I	1.0	1.0	0.0	

In percent values

The number of learners who took the CRLA differs substantially across regions. To enable fair comparisons of performance between regions, we converted raw scores to percentages. This standardization allows us to meaningfully compare aggregate regional performance despite the varying sample sizes.

```
In [15]: %%time
bosy_percentage_df, eosy_percentage_df = crla_analyzer.extract_data_by_period(
    processed_tables,
    data_type='percentages'
)
```

CPU times: user 15 µs, sys: 0 ns, total: 15 µs

Wall time: 18.8 µs

```
In [16]: %%time

combined_percentage_df = crla_analyzer.combine_grade_data(
    bosy_percentage_df,
    eosy_percentage_df,
    school_identifiers
)
```

CPU times: user 84.8 ms, sys: 113 µs, total: 84.9 ms

Wall time: 83.7 ms

```
In [17]: combined_percentage_df['BoSY']
```

Out[17]:

School ID	School Name	Region	Division	District	G1_Developing	G1_Grade Level	G1_Higher Emergent
100000	Balayong Elementary School	Region I	San Carlos City	San Carlos City District I	15.517241	6.896552	12.068966
100001	Apaleng-Libtong ES	Region I	Ilocos Norte	Bacarra I	62.500000	0.000000	0.000000
100002	Bacarra CES	Region I	Ilocos Norte	Bacarra I	12.727273	0.000000	5.454545
100003	Buyon ES	Region I	Ilocos Norte	Bacarra I	0.000000	20.000000	0.000000
100004	Ganagan Elementary School	Region I	Ilocos Norte	Bacarra I	17.647059	0.000000	0.000000
...	...	...	...	...	...	...	...
502889	Maligaya Integrated School	Region III	Science City of Muñoz	Munoz	NaN	NaN	NaN
502890	CURVA Integrated School	Region III	Science City of Muñoz	Munoz	NaN	NaN	NaN
502893	Mayuwi Integrated School	Region IV-A	Tayabas City	Tayabas West	NaN	NaN	NaN
502894	Gonosan Integrated School	Region IX	Zamboanga del Sur	Molave East	NaN	NaN	NaN
502895	Bogo Capalaran Integrated School	Region IX	Zamboanga del Sur	Molave East	NaN	NaN	NaN

37137 rows × 34 columns



In [18]: combined_percentage_df['EoS^Y']

Out[18]:

School ID	School Name	Region	Division	District	G1_Developing	G1_Grade Level	G1_Higher Emergent
100000	Balayong Elementary School	Region I	San Carlos City	San Carlos City District I	10.256410	35.897436	2.564103
100001	Apaleng-Libtong ES	Region I	Ilocos Norte	Bacarra I	14.285714	85.714286	0.000000
100002	Bacarra CES	Region I	Ilocos Norte	Bacarra I	14.545455	41.818182	3.636364
100003	Buyon ES	Region I	Ilocos Norte	Bacarra I	0.000000	53.846154	0.000000
100004	Ganagan Elementary School	Region I	Ilocos Norte	Bacarra I	5.882353	5.882353	0.000000
...	...	...	...	...	...	...	...
502889	Maligaya Integrated School	Region III	Science City of Muñoz	Munoz	14.754098	55.737705	14.754098
502890	CURVA Integrated School	Region III	Science City of Muñoz	Munoz	0.000000	47.500000	0.000000
502893	Mayuwi Integrated School	Region IV-A	Tayabas City	Tayabas West	0.000000	21.052632	0.000000
502894	Gonosan Integrated School	Region IX	Zamboanga del Sur	Molave East	12.500000	0.000000	3.125000
502895	Bogo Capalaran Integrated School	Region IX	Zamboanga del Sur	Molave East	2.857143	88.571429	0.000000

37137 rows × 34 columns



In [19]:

```
%%time
default_analysis_results = crla_analyzer.analyze_reading_performance(
    processed_tables,
    combined_raw_df,
    combined_percentage_df
)
```

CPU times: user 1min 1s, sys: 751 ms, total: 1min 2s
Wall time: 1min 1s

```
In [20]: pca_weights = crla_analyzer.optimize_weights_from_pca()
```

PCA-derived weights based on valid schools:

Developing: 5
Transitioning: 14
Higher Emergent: 9
Lower Emergent: 0
Grade Level: 100

```
In [21]: pca_weights
```

```
Out[21]: {'Developing': 5,  
          'Transitioning': 14,  
          'Higher Emergent': 9,  
          'Lower Emergent': 0,  
          'Grade Level': 100}
```

```
In [22]: %%time  
pca_analysis_results = crla_analyzer.analyze_reading_performance(  
    processed_tables,  
    combined_raw_df,  
    combined_percentage_df,  
    pca_weights  
)
```

Performance calculated using weights: Developing: 5, Transitioning: 14, Higher Emergent: 9, Lower Emergent: 0, Grade Level: 100
CPU times: user 1min 18s, sys: 670 ms, total: 1min 19s
Wall time: 1min 18s

```
In [23]: %%time  
info = crla_analyzer.get_pca_components_info()
```

=== PCA Component Analysis ===
Number of components: 1
Number of features: 30

Component 1 Loadings (Top 5 highest absolute value):
G3_Eng_Grade Level: 0.2846
G3_Fil_Grade Level: 0.2837
G3_MT_Grade Level: 0.2792
G2_MT_Grade Level: 0.2662
G2_Fil_Grade Level: 0.2639

Average Component 1 Loadings by Category:
Grade Level: 0.2684
Lower Emergent: -0.1803
Developing: -0.1572
Higher Emergent: -0.1413
Transitioning: -0.1189

Explained variance ratio: 0.2753
CPU times: user 4.71 ms, sys: 52 μ s, total: 4.76 ms
Wall time: 4.4 ms

```
In [24]: info
```

Out[24]:

G1_Developing **G2_Fil_Developing** **G2_MT_Developing** **G3_Eng_Developing** **G3_Fil_De**

0	-0.109445	-0.159373	-0.158754	-0.175137	.
----------	-----------	-----------	-----------	-----------	---



In [25]:

```
validation = pca_analysis_results['validation']
valid_count = validation['valid_for_progress'].sum()
total_count = len(validation)

print("=== CRLA Data Quality Summary ===")
print(f"Total schools: {total_count}")
print(f"Schools with BoSY data only: {(validation['has_bosy_data'] & ~validation['has_eosy_data']).sum()}")
print(f"Schools with EoSY data only: {(~validation['has_bosy_data'] & validation['has_eosy_data']).sum()}")

if 'count_mismatch' in validation.columns:
    print(f"Schools with significant student count differences: {validation['count_mismatch'].sum()}")
```

```
=== CRLA Data Quality Summary ===
Total schools: 37137
Schools with BoSY data only: 0
Schools with EoSY data only: 0
Schools with significant student count differences: 853
```

In [26]:

```
results = pca_analysis_results['results']
results.head(10)
```

Out[26]:

	School Name	Region	Division	District	Has BoSY Data	Has EoSY Data	Student Count Mismatch	Valid for Progress Analysis	Pc
131072	Gapok Central School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131073	Kapatagan Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131074	Lagubang Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131075	Langgal Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131076	Limuhay Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131077	Matanao Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131078	Paraiso Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131079	Pinandian Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131080	Tacupis Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131081	Don Artemio C. Jacalan Elementary School	Region XII	Sultan Kudarat	Lambayong I	True	True	False	True	

```
In [33]: default_results = default_analysis_results['results']
default_results.head(10)
```

Out[33]:

	School Name	Region	Division	District	Has BoSY Data	Has EoSY Data	Student Count Mismatch	Valid for Progress Analysis	Percentage
131072	Gapok Central School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131073	Kapatagan Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131074	Lagubang Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131075	Langgal Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131076	Limuhay Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131077	Matanao Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131078	Paraiso Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131079	Pinandian Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131080	Tacupis Elementary School	Region XII	Sultan Kudarat	Kulaman II	True	True	False	True	
131081	Don Artemio C. Jacalan Elementary School	Region XII	Sultan Kudarat	Lambayong I	True	True	False	True	

```
In [27]: results[results['Valid for Progress Analysis'] == False].shape
```

Out[27]: (853, 91)

```
In [28]: results[results['Valid for Progress Analysis'] == False].Region.value_counts()
```

```
Out[28]: Region
Region V      179
Region IV-B   130
Region III    73
Region VI     65
CAR           56
Region VIII   52
Region XII    45
Region VII    42
Region I      34
Region IV-A   33
Region II     32
Region XI     31
Region IX     28
Region X      26
CARAGA        17
NCR           6
BARMM         4
Name: count, dtype: int64
```

```
In [29]: results[results['Valid for Progress Analysis'] == True].Region.value_counts()
```

```
Out[29]: Region
Region VIII   3603
Region VI     3360
Region V      2990
Region III    2968
Region VII    2924
Region IV-A   2730
Region I      2380
Region II     2184
Region X      2125
Region IX     2108
Region IV-B   1784
Region XI     1718
Region XII    1703
CARAGA        1691
CAR           1476
NCR           515
BARMM         25
Name: count, dtype: int64
```

```
In [30]: print("=== Top 5 Schools by Progress (Valid Schools Only) ===")
valid_schools = results[results['Valid for Progress Analysis']]
top_progress = valid_schools.sort_values('Progress Score', ascending=False).head(5)
for idx, row in top_progress.iterrows():
    print(f"School ID: {idx}")
    print(f"  School Name: {row['School Name']}")
    print(f"  Region: {row['Region']}")
    print(f"  Progress Score: {row['Progress Score']:.2f}")
    print(f"  BoSY to EoSY Change: {row['BoSY Performance']:.2f} → {row['EoSY Performance']:.2f}")
    print()
```

=== Top 5 Schools by Progress (Valid Schools Only) ===

School ID: 136960

School Name: Mantapay Elementary School

Region: Region XI

Progress Score: 78.12

BoSY to EoSY Change: 0.00 → 78.12

School ID: 100204

School Name: Nagabungan ES

Region: Region I

Progress Score: 76.13

BoSY to EoSY Change: 2.00 → 78.12

School ID: 136975

School Name: Damayon IP School

Region: CARAGA

Progress Score: 75.76

BoSY to EoSY Change: 2.36 → 78.12

School ID: 219515

School Name: Matso-Tokiyas Elementary School (formerly Ambanglo PS)

Region: CAR

Progress Score: 75.52

BoSY to EoSY Change: 2.60 → 78.12

School ID: 135874

School Name: Nagacadan ES

Region: CAR

Progress Score: 74.62

BoSY to EoSY Change: 3.51 → 78.12

```
In [34]: print("=== Top 5 Schools by Progress (Valid Schools Only) - Default Weights ===")
valid_schools = default_results[default_results['Valid for Progress Analysis']]
top_progress = valid_schools.sort_values('Progress Score', ascending=False).head(5)
for idx, row in top_progress.iterrows():
    print(f"School ID: {idx}")
    print(f"  School Name: {row['School Name']}")
    print(f"  Region: {row['Region']}")
    print(f"  Progress Score: {row['Progress Score']:.2f}")
    print(f"  BoSY to EoSY Change: {row['BoSY Performance']:.2f} → {row['EoSY Performance']:.2f}")
    print()
```

=== Top 5 Schools by Progress (Valid Schools Only) - Default Weights ===

School ID: 115320

School Name: Indag-an ES

Region: Region VI

Progress Score: 10.00

BoSY to EoSY Change: 10.00 → 20.00

School ID: 137123

School Name: Cambayang Elementary School

Region: Region IV-B

Progress Score: 6.67

BoSY to EoSY Change: 13.33 → 20.00

School ID: 115775

School Name: Buri PS

Region: Region VI

Progress Score: 6.67

BoSY to EoSY Change: 13.33 → 20.00

School ID: 106501

School Name: Laungcupang ES

Region: Region III

Progress Score: 6.67

BoSY to EoSY Change: 13.33 → 20.00

School ID: 212047

School Name: SIKAT PANLABUAN PS

Region: CARAGA

Progress Score: 6.67

BoSY to EoSY Change: 13.33 → 20.00

```
In [31]: pca_analysis_results['results'].to_csv("data/crla_progress_score.pca_derived.csv")
```

```
In [32]: default_analysis_results['results'].to_csv("data/crla_progress_score.default.csv")
```

Optimizing Reading Proficiency Weights: PCA-Based Analysis

Introduction

Classroom Reading Level Assessment (CRLA) performance scores can be calculated through various weighting methodologies for reading proficiency categories. This analysis employs Principal Component Analysis (PCA) to derive empirically-based weights from assessment data, specifically using only schools with valid data in both assessment periods. The research compares standard unweighted approaches with the data-driven PCA methodology to determine optimal category weighting.

Principal Component Analysis

Mathematical Framework

Principal Component Analysis operates by identifying orthogonal directions of maximum variance in multivariate data. The technique relies on Singular Value Decomposition (SVD), a fundamental matrix factorization method that decomposes the standardized data matrix X into:

$$X = U\Sigma V^T$$

Where:

- U contains left singular vectors (eigenvectors of XX^T)
- Σ contains singular values (square roots of eigenvalues)
- V contains right singular vectors (eigenvectors of X^TX)

The principal components are given by XV , representing projections of the original data onto the eigenvectors.

From Data to Weights: The PCA Process

1. **Standardization:** The data is first standardized using Z-scores: $Z = \frac{X - \mu}{\sigma}$ where X is the original data, μ is the mean, and σ is the standard deviation.
2. **Covariance Matrix:** A covariance matrix C of the standardized data is calculated, which measures how variables change together.
3. **Eigendecomposition:** The eigenvectors and eigenvalues of this matrix are found: $C \cdot v = \lambda \cdot v$ where v is an eigenvector and λ is the corresponding eigenvalue.
4. **Factor Loadings:** The loadings are essentially the correlation between the original variables and the principal components: $L_{ij} = v_{ij} \cdot \sqrt{\lambda_j}$ where L_{ij} is the loading of variable i on component j .
5. **Weight Conversion:** The loadings from the first principal component (the one explaining the most variance) are taken, properly oriented, and scaled to a 0-100 range: $W_i = 100 \cdot \frac{L_i - \min(L)}{\max(L) - \min(L)}$ where W_i is the weight for category i .

The first principal component captures the maximum possible variance, making its loadings particularly valuable as weights. These loadings reflect the relative importance of each reading proficiency category in differentiating school performance based on empirical patterns rather than theoretical assumptions.

Standard vs. PCA-Derived Weights

The comparison between standard unweighted metrics and the PCA-derived weights reveals striking differences:

Reading Proficiency Level	Standard Approach	PCA-Derived Weights
Developing	100	5
Transitioning	100	14
Higher Emergent	100	9
Lower Emergent	100	0
Grade Level	100	100

Key Insights

The PCA-derived weights provide several important insights into reading assessment:

- Hierarchical Importance:** Not all reading proficiency categories contribute equally to performance differentiation. "Grade Level" proficiency appears to be the most discriminating factor in the dataset.
- Counter-Intuitive Contributions:** Despite common assumptions, the "Developing" category (typically considered high proficiency) contributes minimally to differentiation between schools.
- Statistical Redundancy:** The near-zero weight for "Lower Emergent" suggests this category may provide little unique information about school performance beyond what other categories capture.
- Variance Structure:** The relative weights reflect the empirical correlation structure in the data, revealing which proficiency categories actually distinguish between high and low-performing schools.

Methodology Justification

The PCA-based weighting approach is justified for several reasons:

- Data-Driven:** Instead of imposing theoretical weights, the data itself reveals which categories are most important in practice.
- Maximizing Information:** The weights capture the categories that provide the most information about differences between schools.
- Reducing Noise:** By focusing on the strongest patterns, the influence of random variations or measurement errors is minimized.
- Empirical Validation:** The weights reflect actual patterns in student reading performance rather than assumptions about how reading skills should develop.

Implications for Assessment

These findings suggest several important considerations for assessment frameworks:

1. **Equal vs. Differentiated Weighting:** The data strongly suggests that not all reading categories are equally important in explaining differences in school performance, challenging unweighted approaches.
2. **Context-Specific Development:** Reading skill development in the Philippine context may follow different patterns than assumed in standard models, possibly due to language factors, teaching approaches, or cultural contexts.
3. **Assessment Framework Alignment:** The current categories may need recalibration to better reflect how reading skills actually develop in the student population.
4. **Performance Measurement Accuracy:** Using data-derived weights may provide a more accurate reflection of reading proficiency than equal weighting across categories.

Recommendations

Based on these findings, the following recommendations are proposed:

1. **Compare Analysis Results:** Evaluation of how school rankings and performance scores differ between standard and PCA-derived weights to understand the practical impact of these different weighting systems.
2. **Validation Studies:** Consideration of longitudinal studies to determine which weighting system better predicts future academic success or educational outcomes.
3. **Regional and Demographic Analysis:** Investigation into whether optimal weights vary by region, socioeconomic status, or other demographic factors.
4. **Assessment Framework Review:** Subject matter expert review of the current reading proficiency categories in light of these empirical findings.
5. **Mixed Methods Research:** Complementing this quantitative analysis with qualitative research to better understand the contextual factors influencing reading development patterns.

Conclusion

The significant divergence between equal weighting and data-derived weights highlights the importance of empirical validation in educational assessment. While simple models provide valuable frameworks, they should be continually tested against actual learning data. This analysis represents a step toward more evidence-based approaches to measuring and supporting reading development in Philippine schools.

```
# Code snippet showing how PCA weights were derived
pca_weights = crla_analyzer.optimize_weights_from_pca()
print("PCA-derived weights based on valid schools:")
for category, weight in pca_weights.items():
    print(f" {category}: {weight}")
```

These findings invite a reconsideration of understanding reading proficiency assessment and how progress is measured in this critical educational domain.

In []: